

In deep trouble

WE CAN CALL DEEP LEARNING the poster child of research in the domain of artificial intelligence. The interest in deep learning snowballed from the moment the results of the 2012 ImageNet competition declared Geoffrey Hinton's algorithm to have beaten the nearest competitor by 10 percentage points. Ever since then, researchers have been poking and prodding at Convolutional Neural Networks (CNN) and boy has it paid off. In the past six years, the artificial intelligence leaped out of the research lab and became a household name. Most of the gadgets that claim to have AI are simply marketing existing features as AI while some have drawn very broad definitions so that they can fit in with the cool crowd. The truth of the matter is that research in AI has mostly been dominated by deep learning and CNNs are at the heart of it all. And with all such things that get to bask in the glow of mainstream popularity, there will be naysayers who pen articles that spell the doom of said thing. With deep learning, this comes in the form of a new report by the good folks over at the MIT Media Labs.

After having analysed 16,625 research papers from the research database arXiv, it was observed that the deep learning phenomenon has already started slowing down and in the coming years, it could end up on the back burner like many AI domains before it. It's understandable why this is happening. Deep learning based systems have been goofing up quite often and that too in embarrassing ways. All the autonomous vehicles that were involved in accidents were taught using CNNs and CNN-based image classification software such as Amazon's Rekognition ended up with 31 percent false positives in a recent MIT study. The inherent issue with deep learning algorithms is that once the neural network has been trained, there's no way of figuring out how the system is arriving at an answer. There's no chain of 'thought' that you can follow, nor can you reverse engineer the decision making process to figure out the what and the how. This uncertainty is extremely irksome and takes away the aspect of accountability from a system built using deep learning algorithms.

The whole reason we have this field of study is so that we may some day recreate human-like intelligence. To build something with common sense. Deep learning is that kid who gets the right answer most of the times but has no clue how he ended up with that answer. I would never

trust such an entity and many agree with my point of view. That kid could have peaked at the answer, or googled it or asked a buddy or even threatened said buddy with dire consequences to get the answer. I don't know how he got it and that makes me wary. Nevertheless, deep learning has advanced AI research by leaps and bounds. It's like a tool that has served its purpose and needs to be retired. This does not mean that deep learning is dead. In fact, many applications and household gadgets that have become 'smart' thanks to deep learning algorithms are here to stay and more such applications will continue to be developed using CNNs in the years to come.

When it comes to researchers, the domain of study is usually decided by its potential and deep learning seems to be running out of it. The new kid on the block is reinforcement learning, a term used for goal-oriented algorithms which are taught what's the right decision to make on the basis of a reward system. If the algorithm picks whatever you want it to pick, you give it a cookie. Both deep learning and reinforcement learning are autonomous self-learning algorithms. The reward systems helps with one major drawback of deep learning, biases. Since reinforcement learning tries to maximise rewards, if you tell it to not have a race bias, then it'll train itself to meet your conditions. The only problem here is that reinforcement learning relies on constant feedback and if that mechanism has to involve humans, then the rate of learning will be severely limited. However, if you have a well-labelled data set, then you can use it to formulate the right rules and do away with the human feedback element.

The time for deep learning to be in the limelight is certainly nearing the end. Unless someone figures out a way to understand how a deep learning system makes decisions, its days in the research domains are numbered. However, the good thing about AI algorithms is that most of the things we see in action today were actually developed 2-3 decades ago, some going even further back. AI algorithms have a cyclical pattern wherein they shine for a few years and then sulk for a few decades till its time to come out again. Deep learning is on its way into the shadows. We'll still continue to see deep learning applications for a long time to come given how it has helped democratise AI but it's going to be overshadowed by the newest kid on the block. 



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